This is enormously large amount of energy which is enough to keep 100 W bulb operating for 30,000 years. To produce this much energy by chemical reaction, around 20,000 tons of TNT(tri nitro toluene) has to be exploded.

Nuclear reactor:

Nuclear reactor is a device in which the nuclear fission takes place in a self-sustained controlled manner and the energy produced is used either for research purpose or for power generation. The first nuclear reactor was built in the year 1942 at Chicago, USA by physicist Enrico Fermi. The main parts of a nuclear reactor are fuel, moderator and control rods. In addition to this, there is a cooling system which is connected with power generation set up.

Fuel: The fuel is fissionable material, usually uranium or plutonium. Naturally occurring uranium contains only 0.7% of $^{235}_{92}U$ and 99.3% are only $^{238}_{92}U$. So the $^{238}_{92}U$ must be enriched such that it contains at least 2% to 4% of $^{235}_{92}U$. In addition to this, a neutron source is required to initiate the chain reaction for the first time. A mixture of beryllium with plutonium or polonium is used as the neutron source. During fission of $^{235}_{92}U$, only fast neutrons are emitted but the probability of initiating fission by it in another nucleus is very low. Therefore, slow neutrons are preferred for sustained nuclear reactions.

Moderators: The moderator is a material used to convert fast neutrons into slow neutrons. Usually the moderators are chosen in such a way that they must be very light nuclei having mass comparable to that of neutrons. Hence, these light nuclei undergo collision with fast neutrons and the speed of the neutron is reduced (Note that a billiard ball striking a stationary billiard ball of equal mass would itself be stopped but the same billiard ball bounces off almost with same speed when it strikes a heavier mass. This is the reason for using lighter nuclei as moderators). Most of the reactors use heavy water (D_2O) and graphite as moderators. The blocks of uranium stacked together with blocks of graphite (the moderator) to form a large pile is shown in the Figure 9.29 (a) & (b).

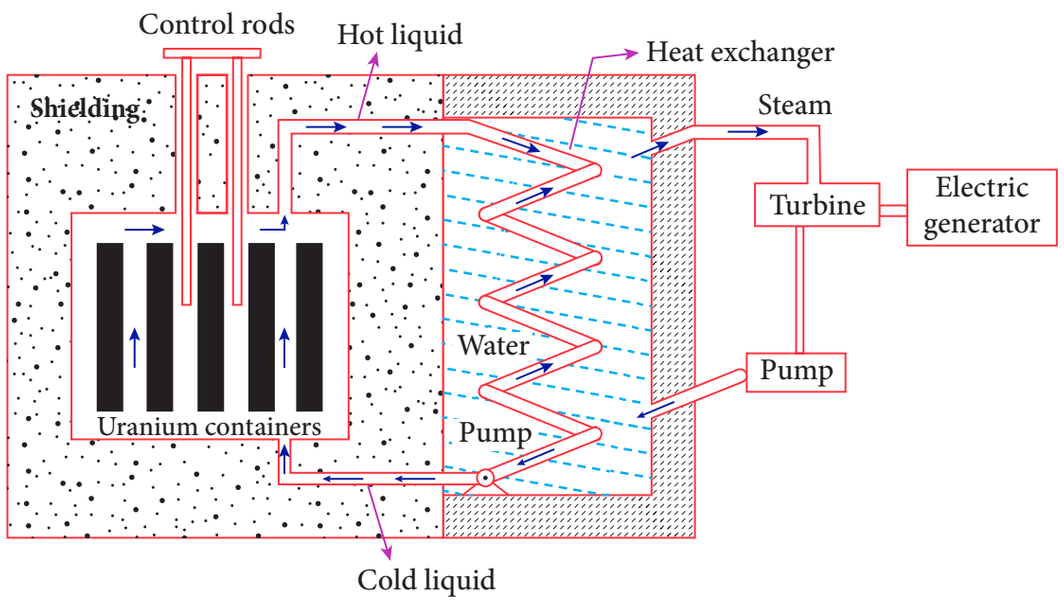


Figure 9.29 (a) Block diagram of Nuclear reactor

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